

1. What is CSS?

CSS (Cascading Style Sheets) is used to style HTML elements.

It controls:

- Layout
- Colors
- Fonts
- Spacing
- Animations

Without CSS → website looks plain

With CSS → website looks professional

2. CSS Syntax

```
selector {  
  property: value;  
}
```

Example:

```
p {  
  color: red;  
}
```

Explanation:

- selector → selects HTML element
- property → what to change
- value → how to change

Use:

To apply style to elements

3. Types of CSS

1. Inline CSS

```
<p style="color: red;">Text</p>
```

Use:

Quick styling (not recommended for large projects)

2. Internal CSS

```
<style>  
p { color: blue; }  
</style>
```

Use:

Styling one page

3. External CSS

```
<link rel="stylesheet" href="style.css">
```

Use:

Best method (used in real projects)

4. CSS Selectors

Universal Selector

```
* { margin: 0; }
```

Selects all elements

Element Selector

```
p { color: blue; }
```

Targets all <p> tags

Class Selector

```
.text { color: green; }
```

Used for multiple elements

ID Selector

```
#header { background: black; }
```

Unique element

Group Selector

```
h1, p { color: red; }
```

Apply same style to multiple elements

5. Colors

```
p { color: red; }  
p { color: #ff0000; }  
p { color: rgb(255,0,0); }
```

Use:

Text and background styling

6. Background Properties

```
body {  
  background-color: lightblue;  
  background-image: url("bg.jpg");  
  background-repeat: no-repeat;  
  background-position: center;  
  background-size: cover;  
}
```

Use:

Design page background

7. Borders

```
div {  
  border: 2px solid black;  
}
```

Use:

Add outline to elements

8. Margin (Outer Space)

```
div {  
  margin: 10px;  
}
```

Use:

Space outside element

9. Padding (Inner Space)

```
div {  
  padding: 10px;  
}
```

Use:

Space inside element

10. Box Model

Structure:

- Content
- Padding
- Border
- Margin

Use:

Layout understanding

11. Width & Height

```
div {  
  width: 200px;  
  height: 100px;  
}
```

Use:

Control element size

12. Text Styling

```
p {  
  color: blue;  
  text-align: center;  
  text-decoration: underline;  
  text-transform: uppercase;  
}
```

Use:

Format text appearance

13. Fonts

```
p {  
  font-family: Arial;  
  font-size: 16px;  
  font-weight: bold;  
}
```

Use:

Improve readability

14. Display Property

```
div { display: block; }  
span { display: inline; }
```

Types:

- block
- inline

- inline-block
- none

Use:

Control layout behavior

15. Position Property

```
div {  
  position: relative;  
  top: 10px;  
}
```

Types:

- static
- relative
- absolute
- fixed
- sticky

Use:

Control element position

16. Flexbox (Important)

```
.container {  
  display: flex;  
  justify-content: center;  
  align-items: center;  
}
```

Use:

Align items easily

17. Grid Layout

```
.container {  
  display: grid;  
  grid-template-columns: repeat(2, 1fr);  
}
```

Use:

Advanced layouts

18. Overflow

```
div {  
  overflow: auto;  
}
```

Use:

Handle extra content

19. Float

```
img {  
  float: right;  
}
```

Use:

Image alignment

20. Z-Index

```
div {  
  z-index: 10;  
}
```

Use:

Control stacking order

21. Opacity

```
div {  
  opacity: 0.5;  
}
```

Use:

Transparency

22. Visibility

```
div {  
  visibility: hidden;  
}
```

Use:

Hide element (space remains)

23. Cursor

```
button {  
  cursor: pointer;  
}
```

Use:

Improve UX

24. Shadows

Box Shadow

```
div {  
  box-shadow: 2px 2px 5px gray;  
}
```

Use:

Visual depth

```
div {  
  background: linear-gradient(red, blue);  
}
```

25. Gradients

Use:

Modern backgrounds

26. Transitions

```
div {  
  transition: 0.5s;  
}
```

Use:

Smooth changes

27. Transform

```
div {  
  transform: scale(1.2);  
}
```

Use:

Resize/rotate elements

28. Animation

```
@keyframes move {  
  from { left: 0px; }  
  to { left: 100px; }  
}  
  
div {  
  animation: move 2s infinite;  
}
```

Use:

Create animations

29. Media Queries (Responsive Design)

```
@media (max-width: 600px) {  
  body {  
    background: red;  
  }  
}
```

Use:

Mobile-friendly design

30. Pseudo Classes

```
a:hover {  
  color: red;  
}
```

Use:

Style based on state

31. Pseudo Elements

```
p::first-letter {  
  font-size: 20px;  
}
```

Use:

Style part of element

32. CSS Units

- px → fixed size
 - % → relative
 - em → relative to parent
 - rem → relative to root
-

33. CSS Variables

```
:root {  
  --main-color: blue;  
}  
  
p {  
  color: var(--main-color);  
}
```

Use:

Reusable values